

## **ENAE464 - 0202**

### **Lab01:** Pressure Measurements of Inviscid Flow over Cylinder

Due on February 13<sup>th</sup>, 2026 at 11:59 AM

*Dr. Silbaugh, 02:00 PM*

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# 1 Abstract

# 2 Nomenclature

# 3 Introduction

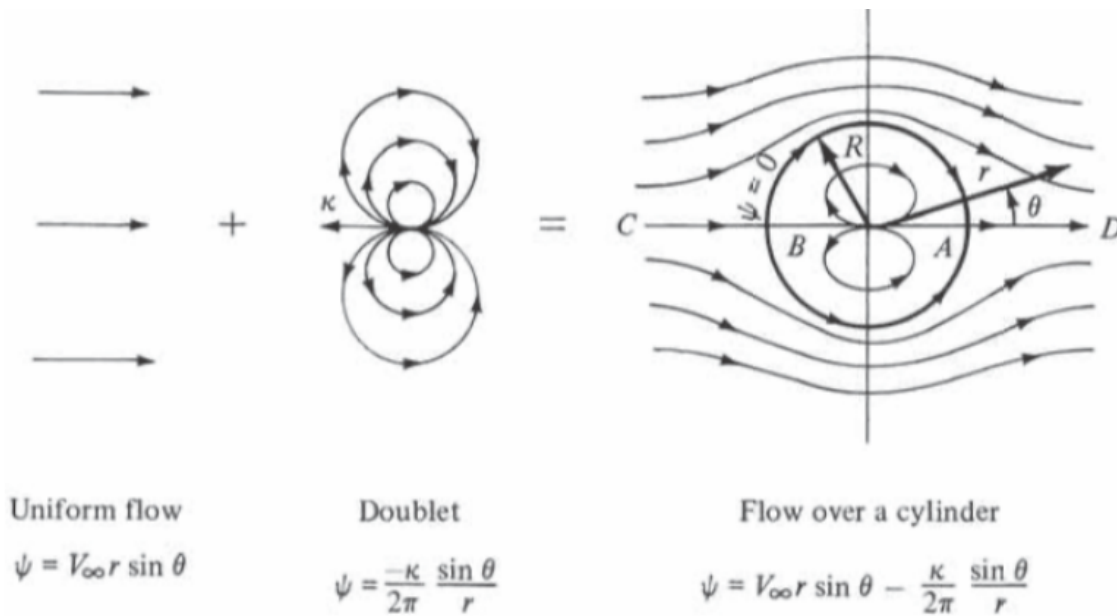
For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):**  $\nabla \cdot \vec{V} = 0$ .
- **Irrotational:**  $\nabla \times \vec{V} = 0$ .

If  $\nabla \times \vec{V} = 0$ , then a *velocity potential*  $\phi$  exists such that  $\vec{V} = \nabla \phi$ . Substituting into continuity gives Laplace's equation:

$$\nabla^2 \phi = 0 \quad \left( \text{i.e., } \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \right).$$

In order to model flow over a 2D cylinder, we use an approach presented in Anderson's Fundamentals of Aerodynamics, which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below:



**Figure 3.26** Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

Figure 1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates  $(r, \theta)$ , the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_{\infty} r \cos(\theta) \quad (\text{velocity potential: uniform flow}) \quad (1)$$

$$\psi_U = V_{\infty} r \sin(\theta) \quad (\text{stream function: uniform flow}) \quad (2)$$

$$\phi_D = \frac{\kappa}{2\pi} \frac{\cos(\theta)}{r} \quad (\text{velocity potential: doublet}) \quad (3)$$

$$\psi_D = -\frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \quad (\text{stream function: doublet}) \quad (4)$$

where  $V_\infty$  is the freestream speed and  $\kappa$  is the doublet strength. Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned} \psi &= \psi_U + \psi_D \\ &= V_\infty r \sin(\theta) - \frac{\kappa}{2\pi} \frac{\sin(\theta)}{r} \\ &= V_\infty r \sin(\theta) \left( 1 - \frac{R^2}{r^2} \right), \end{aligned} \quad (5)$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}.$$

The corresponding velocity field in polar coordinates follows from the stream function relations  $V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$  and  $V_\theta = -\frac{\partial \psi}{\partial r}$ :

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (V_\infty r \cos(\theta)) \left( 1 - \frac{R^2}{r^2} \right) = \left( 1 - \frac{R^2}{r^2} \right) V_\infty \cos(\theta), \quad (6)$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = -\left( 1 + \frac{R^2}{r^2} \right) V_\infty \sin(\theta). \quad (7)$$

At the surface of the cylinder,  $r = R$ , the velocity field is:

$$V_r = \left( 1 - \frac{R^2}{R^2} \right) V_\infty \cos(\theta) = 0, \quad (8)$$

$$V_\theta = -\left( 1 + \frac{R^2}{R^2} \right) V_\infty \sin(\theta) = -2V_\infty \sin(\theta). \quad (9)$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by  $V_r$ . Since  $V_\theta$  is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero,  $V = V_\theta = V_\infty \sin(\theta)$ .

The pressure coefficient is given by:

$$C_p = 1 - \frac{V^2}{V_\infty^2}$$

$$V = -2V_\infty \sin(\theta) \quad (10)$$

$$\frac{V^2}{V_\infty^2} = 4 \sin^2 \theta \quad (11)$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4 \sin^2 \theta} \quad (12)$$

## 4 Experimental Apparatus and Procedures

### 4.1 Experimental Apparatus

### 4.2 Operating Technique

## 5 Results

### 5.1 Operating Conditions

Table 1: Ambient Room Conditions during Experiment

| Quantity              | Value      | Uncertainty  |
|-----------------------|------------|--------------|
| Room Temperature, $T$ | 25 °C      | $\pm 0.5$ °C |
| Room Pressure, $P$    | 1009.0 hPa | –            |

The density of air ( $\rho$ ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT}$$

where:

- $P$  is the absolute pressure (in SI units: Pa)
- $T$  is the absolute temperature (in Kelvin)
- $R$  is the specific gas constant for dry air,  $R = 287.05 \text{ J}/(\text{kg} \cdot \text{K})$

Given measurements:

$$P = 1009.0 \text{ hPa} = 100900 \text{ Pa}$$

$$T = 25 \text{ °C} = 25 + 273.15 = 298.15 \text{ K}$$

$$R = 287.05 \text{ J}/(\text{kg} \cdot \text{K})$$

$$\rho = \frac{100900 \text{ Pa}}{287.05 \text{ J}/(\text{kg} \cdot \text{K}) \times 298.15 \text{ K}}$$

Calculate:

$$\begin{aligned} \rho &= \frac{100900}{287.05 \times 298.15} \\ &= \frac{100900}{85542.5} \\ &\approx 1.18 \text{ kg/m}^3 \end{aligned}$$

Thus, the ambient air density during the experiment is:

$$\boxed{\rho = 1.18 \text{ kg/m}^3}$$

## 5.2 Measurements

## 5.3 Quantities of Interest

# 6 Analysis and Discussion

## 6.1 Data Conversion

## 6.2 Interpretation

# 7 Summary and Conclusions

# 8 References

## References

- [1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, pp. 209–324, ISBN: 9781266076442.

## 9 Appendices

### 9.1 Data

| ./data/pressure_vs_alpha.csv |              |                   |            |          |
|------------------------------|--------------|-------------------|------------|----------|
|                              | $\alpha$ [°] | $P_{\infty}$ [cm] | $P_0$ [cm] | $P$ [cm] |
| 01                           | 0            | 15.4              | 23.0       | 23.0     |
| 02                           | 5            | 15.4              | 23.1       | 22.7     |
| 03                           | 10           | 15.4              | 23.1       | 22.3     |
| 04                           | 15           | 15.4              | 23.2       | 21.3     |
| 05                           | 20           | 15.3              | 23.1       | 19.9     |
| 06                           | 25           | 15.4              | 23.1       | 18.6     |
| 07                           | 30           | 15.4              | 23.2       | 16.9     |
| 08                           | 35           | 15.4              | 23.1       | 15.0     |
| 09                           | 40           | 15.4              | 23.1       | 13.1     |
| 10                           | 45           | 15.4              | 23.1       | 11.2     |
| 11                           | 50           | 15.3              | 23.1       | 9.4      |
| 12                           | 55           | 15.3              | 23.1       | 8.4      |
| 13                           | 60           | 15.4              | 23.1       | 7.4      |
| 14                           | 65           | 15.4              | 23.1       | 7.0      |
| 15                           | 70           | 15.4              | 23.2       | 6.9      |
| 16                           | 75           | 15.4              | 23.2       | 7.6      |
| 17                           | 80           | 15.4              | 23.2       | 7.8      |
| 18                           | 85           | 15.4              | 23.1       | 8.0      |
| 19                           | 90           | 15.4              | 23.1       | 8.1      |
| 20                           | 95           | 15.4              | 23.1       | 8.3      |
| 21                           | 100          | 15.4              | 23.2       | 8.3      |
| 22                           | 105          | 15.4              | 23.1       | 8.2      |
| 23                           | 110          | 15.4              | 23.1       | 8.1      |
| 24                           | 115          | 15.4              | 23.1       | 8.1      |
| 25                           | 120          | 15.4              | 23.1       | 8.1      |
| 26                           | 125          | 15.4              | 23.1       | 7.9      |
| 27                           | 130          | 15.4              | 23.1       | 7.8      |
| 28                           | 135          | 15.4              | 23.1       | 7.8      |
| 29                           | 140          | 15.4              | 23.1       | 7.7      |
| 30                           | 145          | 15.4              | 23.1       | 7.6      |
| 31                           | 150          | 15.4              | 23.1       | 7.7      |
| 32                           | 155          | 15.4              | 23.1       | 7.8      |
| 33                           | 160          | 15.4              | 23.0       | 8.0      |
| 34                           | 165          | 15.4              | 23.1       | 8.0      |
| 35                           | 170          | 15.4              | 23.1       | 8.2      |
| 36                           | 175          | 15.4              | 23.1       | 8.2      |
| 37                           | 180          | 15.4              | 23.1       | 8.2      |

Table 2: Raw Measurements of Pressure Taps vs. Angle of Attack



## 9.2 Code